

mRNA Model

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1 Polymer Conformational Model

We approximate the mRNA as a freely jointed chain (FJC):

$$\rho = \sum_{i=1}^N \mathbf{b}_i, \quad |\mathbf{b}_i| = b,$$

where b is the Kuhn length and $N = L/b$ is the number of Kuhn segments.

For sufficiently large N , the end-to-end vector can be approximated by a Gaussian distribution:

$$p(\rho) = \left(\frac{3}{2\pi N b^2} \right)^{3/2} \exp \left(-\frac{3\rho^2}{2N b^2} \right)$$

where $p(\rho)$ is the three-dimensional probability density of the end-to-end vector and $\rho = |\rho|$.

Since the protein-mediated bridge depends primarily on the distance between the two interaction sites, we integrate over all orientations to obtain the radial probability density:

$$p_\rho(\rho) = 4\pi\rho^2 p(\rho)$$

or explicitly,

$$p_\rho(\rho) = 4\pi\rho^2 \left(\frac{3}{2\pi N b^2} \right)^{3/2} \exp \left(-\frac{3\rho^2}{2N b^2} \right).$$

We assume that protein-mediated bridging is geometrically possible when the distance between the two interaction sites lies within a capture range:

$$\rho_{\min} \leq \rho \leq \rho_{\max}.$$

The equilibrium fraction of RNA conformations within this protein-compatible range is therefore

$$P_{\text{enc}}^{\text{loop}} = \int_{\rho_{\min}}^{\rho_{\max}} p_\rho(\rho) d\rho$$

where $P_{\text{enc}}^{\text{loop}}$ is dimensionless.

The corresponding capture region is a spherical shell with volume

$$V_{\text{loop}} = \frac{4\pi}{3} (\rho_{\text{max}}^3 - \rho_{\text{min}}^3)$$

We define the average spatial probability density within this capture shell as

$$\bar{p}_{\text{loop}} = \frac{P_{\text{enc}}^{\text{loop}}}{V_{\text{loop}}}$$

which has units of inverse volume.

To convert this spatial probability density into an effective concentration, we use the relation between the spatial density of one molecule and its molar concentration:

$$C_{\text{eff}}^{\text{loop}} = \frac{\bar{p}_{\text{loop}}}{N_A},$$

with the appropriate unit conversion between volume and molarity.

Thus,

$$C_{\text{eff}}^{\text{loop}} = \frac{1}{N_A} \frac{\int_{\rho_{\text{min}}}^{\rho_{\text{max}}} 4\pi \rho^2 p(\rho) d\rho}{\frac{4\pi}{3} (\rho_{\text{max}}^3 - \rho_{\text{min}}^3)}$$

where $C_{\text{eff}}^{\text{loop}}$ represents the average effective concentration of one interaction site around the other within the protein-compatible capture shell.

2 Protein-Mediated Bridging

We next consider protein-mediated bridging between the 5' and 3' ends of the mRNA, for example through interactions involving PABP and eIF4G.

The standard-state bimolecular association constant is

$$K_{\text{bind}}^{\circ} = \frac{1}{C^{\circ}} \exp\left(-\frac{\Delta G_{\text{bind}}^{\circ}}{RT}\right)$$

where $C^{\circ} = 1 \text{ M}$ is the standard concentration.

The intramolecular looping equilibrium is then described by

$$K_{\text{loop}} = K_{\text{bind}}^{\circ} C_{\text{eff}}^{\text{loop}}$$

or equivalently,

$$K_{\text{loop}} = \frac{C_{\text{eff}}^{\text{loop}}}{C^{\circ}} \exp\left(-\frac{\Delta G_{\text{bind}}^{\circ}}{RT}\right)$$

Finally, treating the mRNA as a two-state system,

$$\text{Open} \rightleftharpoons \text{Closed},$$

the equilibrium fraction of closed-loop conformations is

$$f_{\text{closed}} = \frac{K_{\text{loop}}}{1 + K_{\text{loop}}}$$

3 Closed-Loop Positional Model

Once the mRNA adopts the closed-loop state, we approximate the mRNA as a Gaussian ring and consider the spatial encounter between the BD and the 5' translation-initiation machinery.

Let n denote the number of Kuhn segments along one contour path between the BD and the 5' translation-initiation anchor. The complementary contour path therefore contains $N - n$ Kuhn segments. The spatial separation between the two sites is denoted by R .

For a Gaussian ring, the three-dimensional probability density of the spatial separation R is

$$p_{\text{ring}}(\mathbf{R} \mid n) = \left[\frac{3N}{2\pi b^2 n(N - n)} \right]^{3/2} \exp \left[-\frac{3NR^2}{2b^2 n(N - n)} \right]$$

where $R = |\mathbf{R}|$.

Integrating over all orientations gives the corresponding radial probability density:

$$p_{\text{ring}}(R \mid n) = 4\pi R^2 \left[\frac{3N}{2\pi b^2 n(N - n)} \right]^{3/2} \exp \left[-\frac{3NR^2}{2b^2 n(N - n)} \right]$$

Because the interaction between the BD and the 5' translation-initiation machinery involves finite-sized molecular interfaces rather than an ideal point contact, we assume that productive interaction occurs when their spatial separation lies within a finite capture range,

$$R_{\text{min}} \leq R \leq R_{\text{max}}.$$

The probability of finding the BD within this capture range is therefore

$$P_{\text{enc}}^{\text{ring}}(n) = \int_{R_{\text{min}}}^{R_{\text{max}}} p_{\text{ring}}(R \mid n) dR$$

with the corresponding capture-shell volume

$$V_{\text{BD}} = \frac{4\pi}{3} (R_{\text{max}}^3 - R_{\text{min}}^3)$$

We define the shell-averaged effective concentration of the BD around the 5' translation-initiation machinery as

$$C_{\text{eff}}^{\text{ring}}(n) = \frac{1}{N_A} \frac{P_{\text{enc}}^{\text{ring}}(n)}{V_{\text{BD}}}$$

with the appropriate unit conversion between volume and molarity. Equivalently,

$$C_{\text{eff}}^{\text{ring}}(n) = \frac{1}{N_A V_{\text{BD}}} \int_{R_{\text{min}}}^{R_{\text{max}}} 4\pi R^2 \left[\frac{3N}{2\pi b^2 n(N-n)} \right]^{3/2} \exp \left[-\frac{3NR^2}{2b^2 n(N-n)} \right] dR$$

This quantity describes the positional contribution of the polymer conformation to the interaction between the BD and the 5' translation-initiation machinery.

3.1 cis Configuration

In the cis configuration, the BD is covalently incorporated into the same mRNA molecule as the 5' translation-initiation machinery. Therefore, no additional intermolecular BD–RNA binding free energy is required.

The translational effect is consequently determined by the probability of forming the closed-loop state and the positional effective concentration of the BD:

$$TE_{\text{cis}}(n) \propto f_{\text{closed}} C_{\text{eff}}^{\text{ring}}(n)$$

Thus, under the assumption that the presence of the BD does not substantially perturb the open–closed equilibrium, the positional dependence of the cis configuration is determined by

$$TE_{\text{cis}}(n) \propto C_{\text{eff}}^{\text{ring}}(n)$$

where n specifies the position of the BD along the closed-loop mRNA.

3.2 trans Configuration

In the trans configuration, the HEPHA-RNA carrying the BD is an independent RNA molecule. In addition to the spatial encounter between the BD and the 5' translation-initiation machinery, productive regulation therefore requires binding of the BD to its target site on the mRNA.

The standard-state association constant for BD–target RNA binding is

$$K_{\text{BD}}^{\circ}(n) = \frac{1}{C^{\circ}} \exp \left[-\frac{\Delta G_{\text{BD}}^{\circ}(n)}{RT} \right]$$

where $C^{\circ} = 1 \text{ M}$ is the standard concentration and $\Delta G_{\text{BD}}^{\circ}(n)$ is the standard binding free energy between the BD and the target site at position n .

Importantly, $\Delta G_{\text{BD}}^{\circ}(n)$ is explicitly position-dependent because different target sites along the mRNA may have different sequence and structural properties, leading to different binding affinities.

The spatial encounter and molecular binding contributions can then be combined into an effective intramolecular association term:

$$K_{\text{trans}}(n) = K_{\text{BD}}^{\circ}(n) C_{\text{eff}}^{\text{ring}}(n)$$

or equivalently,

$$K_{\text{trans}}(n) = \frac{C_{\text{eff}}^{\text{ring}}(n)}{C^{\circ}} \exp \left[-\frac{\Delta G_{\text{BD}}^{\circ}(n)}{RT} \right]$$

This expression separates the two contributions to trans-acting regulation: $C_{\text{eff}}^{\text{ring}}(n)$ describes the positional and polymer-conformational contribution, whereas $\Delta G_{\text{BD}}^{\circ}(n)$ describes the sequence- and structure-dependent binding contribution.

Consequently, the translational effect of a trans-acting BD targeting site n can be written as

$$TE_{\text{trans}}(n) \propto f_{\text{closed}} C_{\text{eff}}^{\text{ring}}(n) K_{\text{BD}}^{\circ}(n)$$

or

$$TE_{\text{trans}}(n) \propto f_{\text{closed}} \frac{C_{\text{eff}}^{\text{ring}}(n)}{C^{\circ}} \exp \left[-\frac{\Delta G_{\text{BD}}^{\circ}(n)}{RT} \right]$$

Thus, the model naturally decomposes the positional effect of a trans-acting element into two independent contributions: the spatial accessibility determined by polymer conformational statistics and the binding affinity determined by the free energy of BD–target RNA interaction.

For comparing different target sites on the same mRNA, f_{closed} is a common factor and can therefore be omitted. The site-dependent contribution to trans-acting regulation is then

$$TE_{\text{trans}}(n) \propto C_{\text{eff}}^{\text{ring}}(n) K_{\text{BD}}^{\circ}(n)$$

or equivalently,

$$TE_{\text{trans}}(n) \propto \frac{C_{\text{eff}}^{\text{ring}}(n)}{C^{\circ}} \exp \left[-\frac{\Delta G_{\text{BD}}^{\circ}(n)}{RT} \right]$$

Thus, when comparing candidate BD binding sites, the model contains two site-dependent contributions:

$$\text{Sitescore}(n) \propto \underbrace{C_{\text{eff}}^{\text{ring}}(n)}_{\text{polymerpositionalcontribution}} \underbrace{\exp \left[-\frac{\Delta G_{\text{BD}}^{\circ}(n)}{RT} \right]}_{\text{bindingcontribution}}$$

The first term captures the positional accessibility of the target site within the closed-loop mRNA, whereas the second term captures the sequence- and structure-dependent affinity of the BD for that site.